

Homotopy Invariance and the Thom Isomorphism from Stokes' Theorem

Claudio Bastiani-Fonck

1. Introduction

This note is meant to give a relatively geometrical argument for homotopy invariance as well as the Thom isomorphism in DeRham cohomology. It was made as part of my reading course with Aaron Calderon on differential forms in Algebraic Topology. I have tried to include a wide variety of visuals, and have generally preferred intuition over rigor, so one should probably already believe the results are true before reading this.

2. The Proof

Theorem 2.1 (Homotopy Invariance).

$$H^k(M \times \mathbb{R}) \cong H^k(M)$$

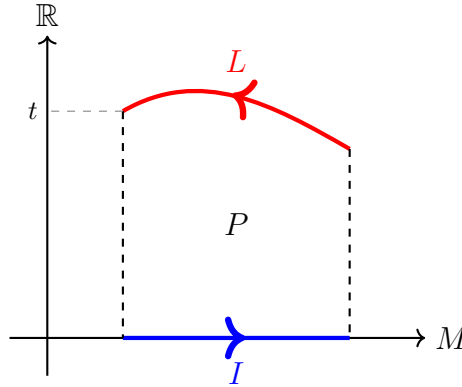
Proof. Let $\pi : M \times \mathbb{R} \rightarrow M$ be the standard projection and $i : M \rightarrow M \times \mathbb{R}$ with $i(p) = (p, 0)$ be the inclusion to 0. We can consider the diagram

$$\begin{array}{ccc} M \times \mathbb{R} & & \\ i \uparrow & \searrow \pi & \\ M & & \end{array}$$

and claim that π and i are mutually inverse isomorphisms in cohomology. One direction is clear:

$$\pi \circ i = \text{id} \implies i^* \circ \pi^* = \text{id}$$

Therefore we must focus on $\pi^* \circ i^*$. We need to show that for any closed form $\omega \in \Omega(M \times \mathbb{R})$ that $\tau := \omega - (\pi^* \circ i^*)(\omega)$ is an exact form. We can understand τ by considering what happens when we integrate it along some well behaved region L which looks like a graph.



We let $I := i \circ \pi(L)$ be the projection of L down to $t = 0$ and define

$$P := \{(p, t) \in M \times \mathbb{R} \mid 0 \leq t/t' \leq 1 \text{ for some } (p, t') \in L\}$$

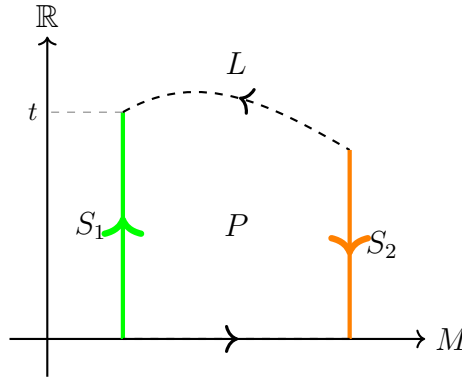
to be the *prism* contained between L and I . We give L and I the orientation induced on the boundary of P . The integral of $(\pi^* \circ i^*)(\omega)$ along L is the integral of ω along I with the orientation induced by the map. This is the opposite of the orientation we are using so:

$$\int_L \tau = \int_L \omega - \int_L (\pi^* \circ i^*)(\omega) = \int_L \omega + \int_I \omega \quad (1)$$

We seek to find a form whose exterior derivative is τ . We might try to "sweep over" the difference between L and I with the following operator

$$K(\alpha \wedge dt)(p, t) := \int_0^t \alpha(p, s) ds$$

which takes a differential form and integrates it along $\mathbb{R}$ from 0 to the point of evaluation. I claim that $\tau = d(K\omega)$, to see this we will consider the integral of $d(K\omega)$ along L .



Define

$$S := \{(p, t) \in M \times \mathbb{R} \mid 0 \leq t/t' \leq 1 \text{ for some } (p, t') \in \partial L\}$$

to be the "side" of P . By Stokes' theorem

$$\begin{aligned} \int_L dK(\omega) &= \int_{\partial L} K(\omega) \\ &= \int_{(p,t) \in \partial L} \int_{s=0}^t \omega(p, s) \\ &= - \int_S \omega \end{aligned} \quad (2)$$

which corresponds to the integral of ω along the colored segments in the above diagram. By Stokes' theorem

$$\int_I \omega + \int_L \omega + \int_S \omega = \int_{\partial P} \omega = 0 \quad (3)$$

since ω is closed. This tells us exactly that

$$\int_L \tau = \int_L d(K\omega)$$

after plugging in Eq. (1) and Eq. (2). Since L was arbitrary some surprisingly heavy machinery implies that $\tau = d(K\omega)$. $\square$

Remark 2.2. The standard proof usually utilizes the more general fact that

$$\tau = K(d\omega) + d(K\omega)$$

which is true even on non exact forms. To see why this is true, note that

$$\begin{aligned}
\int_L K(d\omega) &= \int_{(p,t) \in L} \int_{s=0}^t d\omega(p,s) \\
&= \int_P d\omega \\
&= \int_{\partial P} \omega
\end{aligned}
\qquad \text{Stokes' theorem}$$

so the contribution from this term exactly accounts for what we get on the RHS of Eq. (3) when ω is not closed.

The next theorem nicely wraps up all the work we have done into a very strong and conceptual result which is useful for understanding/reasoning about the subject as well as solving problems.

Theorem 2.3. *Let $f, g : M \rightarrow N$ be smooth maps which can be smoothly deformed to each other. Then f and g induce the same map on cohomology ($f^* = g^*$).*

Proof. The statement that f and g can be continuously deformed to each other is that we can find a map:

$$H : M \times I \rightarrow N \quad H(p, 0) = f(p) \quad H(p, 1) = g(p) \quad I := [0, 1] \subset \mathbb{R}$$

Therefore write $H \circ i_0 = f$ and $H \circ i_1 = g$, where $i_t : M \rightarrow \mathbb{R} \times M$ is the inclusion $x \mapsto (t, x)$ and define $\pi : M \times \mathbb{R} \rightarrow M$ to be the standard projection. Observe the relationship between the inclusions and the projection on the manifold level:

$$\begin{array}{ccccc}
& & i_0 & & \\
& \nearrow & & \searrow & \\
M & \xleftarrow{\pi} & M \times I & \xrightarrow{H} & N \\
& \searrow & & \nearrow & \\
& & i_1 & &
\end{array}$$

Note that our proof of Theorem 2.1 works just as well to show that

$$H^k(M \times I) \cong H^k(M)$$

so on the level of cohomology i_0^* and i_1^* are both the unique inverse to π^* , therefore the same map. Then the compositions $f^* = i_0^* \circ H^*$ and $g^* = i_1^* \circ H^*$ are the same map on cohomology as desired. $\square$

Corollary 2.4. *Any contractible manifold has $H^0 = \mathbb{R}$ and $H^k = 0$ for $k \neq 0$.*

Theorem 2.5 (Thom Isomorphism). *Suppose that E is the total space of a vector bundle of rank n over the base space M . If $H_{cv}^k(E)$ is the cohomology of vertically compactly supported forms then:*

$$H_{cv}^{k-n}(E) \cong H^k(M)$$

Proof. By the Mayor-Vietoris argument we may work locally where the vector bundle is trivial [2]. Therefore assume $E \cong M \times \mathbb{R}^n$. We also initially assume that $n = 1$. Define $e(t)$ as a bump function on $\mathbb{R}$ that is compactly supported around 0 and integrates to 1. Now consider the maps

$$\begin{array}{c}
\Omega_{cv}(M \times \mathbb{R}) \\
\begin{array}{c} \nearrow \quad \searrow \\ \rho \quad I \end{array} \\
\Omega(M)
\end{array}$$

where I is the map that integrates a form along the fiber $\mathbb{R}$ and ρ is the map that wedges the bump function:

$$I(\alpha \wedge dt)(p) := \int_{-\infty}^{\infty} \alpha(p, s) ds \quad \rho(\alpha) := e(t) dt \wedge \alpha$$

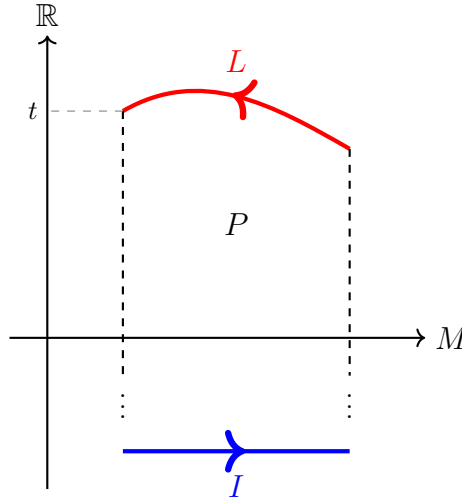
Essentially I integrates out or averages along the vertical direction while ρ takes a form on M and smears or spreads it out along the vertical direction. I claim that I and ρ induce inverse isomorphisms in cohomology. One immediately sees

$$(I \circ \rho)(\omega) = \omega \int_{-\infty}^{\infty} e(t) dt = \omega$$

by our construction. Therefore we need to show for some closed form $\omega \in \Omega_{cv}(M \times \mathbb{R})$ that $\tau := \omega - (\rho \circ I)(\omega)$ is exact. For this we define the operator

$$K(\alpha \wedge dt)(p, t) := \int_{-\infty}^t \alpha(p, s) ds - E(t) \int_{-\infty}^{\infty} \alpha(p, s) ds \quad E(t) := \int_{-\infty}^t e(s) ds$$

and claim that $d(K\omega)$ is τ . Note that the first term from $\int_{-\infty}^t \alpha(p, s) ds$ is very similar to what we encountered in the proof of Theorem 2.1. However now we are integrating from $-\infty$ instead of 0.



By the same argument as in that proof, we obtain

$$\int_L dK(\omega) = \int_L \omega + \int_I \omega$$

in the limit as we push I to $t = -\infty$. Since ω is vertically compactly supported, this kills the integral along I and

$$\int_L dK(\omega) = \int_L \omega$$

so this term accounts for ω . Therefore the second term $-E(t) \int_{-\infty}^{\infty} \alpha(p, s) ds$ should account for $(\rho \circ I)(\omega) = e(t)dt \wedge \tilde{\omega}$ (where $\tilde{\omega} := I(\omega)$). Applying the second term to ω and differentiating directly yields

$$d(E(t)\tilde{\omega}) = e(t)dt \wedge \tilde{\omega} + E(t)d\tilde{\omega}$$

and we are reduced to proving that $d\tilde{\omega} = 0$. This comes from the fact that ω is compactly vertically supported and we don't have a boundary to worry about (allowing us to commute the exterior derivative and integral):

$$d\tilde{\omega} = d \int_{s=-\infty}^{\infty} \omega(p, s) = \int_{s=-\infty}^{\infty} d\omega(p, s) = 0$$

We have used the fact that $d\omega = 0$ since ω is closed.

Now we would like to extend this to the case when $n \neq 1$. We do this by integrating out the fiber dimensions one by one, supposing the Thom isomorphism works for $n \leq r$.

$$\begin{array}{c}
 \Omega_{cv}(M \times \mathbb{R}^{r+1}) \\
 \rho_1 \left(\begin{array}{c} \nearrow \\ \searrow \end{array} \right) I_1 \\
 \Omega_{cv}(M \times \mathbb{R}^r) \\
 \rho \left(\begin{array}{c} \nearrow \\ \searrow \end{array} \right) I \\
 \Omega(M)
 \end{array}$$

□

We have denoted I_1 to be the integration along one fiber direction t_1 and ρ_1 to be wedging by a bump function along this direction. Since being vertically compactly supported can be checked on each dimension separately, the induction goes through.

References

- [1] Aaron Calderon, *Reading Course*, UChicago Winter Quarter 2026.
- [2] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Graduate Texts in Mathematics, Vol. 82, Springer-Verlag, New York, 1982.